Text Preprocessing

Text Data → Sentences, documents, etc.

Ex- Email : "You have a meeting tomorrow."

Review : "The colour of the shirt is beautiful".

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For this data, normal data preprocessing does not work well.

So, for such data, we use Text Preprocessing.

Ques. Can we encode Text Data using categorical encoding techniques?

Ans. No, these encoding techniques cannot be used in case of Text data.

Ques. How Text Preprocessing is done?

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Text Preprocessing Steps :-

① Text Cleaning :- Removing unnecessary punctuations, spaces, special characters, etc. from the data.

Ex- Meeting tomorrow @ 5p.m. ⇝ uncleaned data

Meeting tomorrow 5pm ⇝ cleaned data

② Tokenization :- Breaking the text into smaller units.

a) Word Tokenization :- The Text is splitted into individual words.

Ex- "Meeting tomorrow at 5pm" ⇝ ["Meeting", "tomorrow", "at",

"5", "pm"]

b) Sentence Tokenization :- The Text is splitted into sentences.

Ex- "I love data science. It is my favourite subject."

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["I love data science.", "It is my favourite subject"]

(3) Stopwords Removal :-

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common but unimportant words.

Ex- to, the, it, or, that, in, as, etc.

(4) Stemming / Lemmatization :-

a) Stemming :- Reducing the word to their root form.

Ex- running → runn

changing → chang

studies → studi

b) Lemmatization :- Reducing the word to their base form.

Ex- changing → change

studies → study

running → run

(5) Encoding the text data :- Mulitple techniques are used for converting text into nos.

Ex- Bag of words (BOW), TF-IDF, Word 2 Vec, etc.

* Bag of Words (BOW) :- [Binary BOW]

- A text representation technique used to convert text data into numerical data.
- It represents a piece of text as a collection of bag of words.

It works in 3 steps :-

a) Vocabulary Creation
b) Vector Creation
c) Document / Sentence Representation.

Ex- Let us take 3 sentences

S1 → I like pizza.

S2 → I do not like pizza.

S3 → I like to play cricket.

(1) Vocabulary Creation :- [Using Unique Words]

["I", "like", "pizza", "do", "not", "to", "play", "cricket"]

Vocabulary / corpus

So, Total Sentences = 3

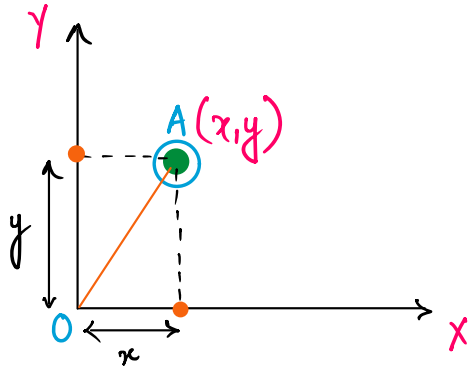
Total Words = 8

Total Words = 8

(2) Vector Creation :-

Vector → An array of nos.

Vector A = [x, y]



Sentences	I	like	pizza	do	not	to	play	cricket
I like Pizza	1	1	1	0	0	0	0	0
I do not like Pizza	1	1	1	1	1	0	0	0
I like to play cricket	1	1	0	0	0	1	1	1

(3) Sentence Representation :-

S1 → [1 1 1 0 0 0 0 0]

S2 → [1 1 1 1 1 0 0 0]

S3 → [1 1 0 0 0 1 1 1]

But what if in my sentence, a particular word occurs more than 1 time ?

Ans → Count BOW

Ex- I like pizza because pizza is tasty.

Ex- I like pizza because pizza is tasty.

Vocab →	I	like	pizza	because	is	tasty
Binary BOW →	1	1	1	1	1	1
Count BOW →	1	1	2	1	1	1

* Drawbacks of BOW :-

1. No order of words is maintained.
2. High Dimensionality
3. Sparsity
4. Equal importance to all the words